NarrateIQ

AIStory-BasedLearning for childrens and Specially-Abled Children

EdTech | AI/ML | Accessibility | Inclusive Education | ISL | Multimodal

Domain	Inclusive Education & Assistive Technology
Project Theme	AI-Powered Multimodal Learning for Differently-Abled Children
Project Type	Software Platform (Web + Mobile)
Target Beneficiaries	Children with Dyslexia, Autism, ADHD, Hearing Impairment Paired

■ PROBLEM STATEMENT

India has over **26 million children** with some form of disability, of which a significant proportion face learning disabilities such as Dyslexia, Autism Spectrum Disorder (ASD), and ADHD. Despite the Right to Education Act guaranteeing inclusive education, the reality on the ground is starkly different.

One-Size-Fits-All Content	Standard textbooks and digital content are not designed for differently-abled learners. A child with dyslexia processes text differently than one with autism or ADHD.
No Multilingual Accessibility	Special education content is overwhelmingly in English, leaving millions of children in regional-language households without suitable resources.
Shortage of Special Educators	India has a deficit of over 1.2 million trained special educators. Teachers in mainstream schools are not equipped to create personalized learning materials.
Hearing-Impaired Children Left Behind	Over 6.3 lakh children with hearing impairment have virtually no access to Indian Sign Language (ISL)-based academic content — a critical gap in educational equity.
No Real-Time Adaptive System	Existing EdTech platforms lack dynamic adaptation to the learning pace and disability type of individual children, making them ineffective for special needs education.

■ PROPOSED SOLUTION — NarrateIQ

NarrateIQ is an AI-powered, multimodal adaptive learning platform that transforms any educational topic into personalized, accessible learning experiences tailored to each child's specific disability type — delivered as audio stories, animated visual explanations, and Indian Sign Language (ISL) avatar videos.

HOW IT WORKS — End-to-End Flow

STEP	ACTION	OUTPUT
Teacher	Enters topic via Text / Audio / Video upload	Structured topic + context
AI Engine	Detects disability type profile selected	Personalization parameters
Story Gen	LLM generates age-appropriate narrative	Engaging story in local language
Audio TTS	Converts story to speech in regional dialect	MP3 audio story
Visual AI	Generates animated visual scenes	MP4 animated explanation
ISL Engine	Renders ISL avatar signing the content	ISL video output
Dashboard	Teacher/Parent reviews & assigns to child	Personalized content card

▪ TECHNOLOGY STACK

Layer	Technology	Purpose
Frontend	React.js + Flutter	Web & Mobile App (Teacher + Parent)
Input Processing	Whisper STT + OpenCV	Audio/Video transcription & scene extract
NLP / Story Gen	GPT-4o + LangChain + IndicBERT	Adaptive story generation per disability
TTS / Audio	Bhashini API + Coqui TTS	Regional language audio narration
Animation	Manim + Stable Diffusion	Animated visual scene generation Real-
ISL Engine	Custom ISL GAN + MediaPipe	time ISL avatar video rendering Async
Backend	FastAPI + Celery + Redis	processing & task queuing Relational data
Database	PostgreSQL + Pinecone	+ vector embeddings Scalable, govt-
Cloud / Deploy	AWS / NIC Cloud + Docker	compliant deployment